constants

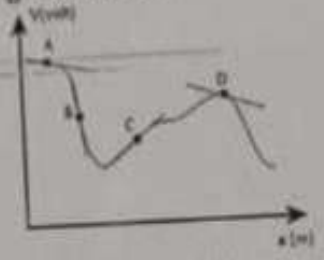
$$q_e = -1.6\times10^{-19}C, q_p = +1.6\times10^{-19}C, m_e = 9.11\times10^{-31}kg, m_p = 1.672\times10^{-27}kg,$$
$$\epsilon_0 = 8.85\times\frac{10^{-12}C^2}{N\cdot m^2}, k_e = 8.987\times\frac{10^9 N\cdot m^2}{C^2}$$

يتكون الجزء الأول من 16 سؤال إختيار من متعدد، بعد اختيار رمز الإجابة الصحيحة يتم تلبينها في الجدول المخصص بالصفحة الأولى

Q.1) A proton is released from rest in a uniform electric field that has a magnitude of 6.0×10^4 V/m. The proton undergoes a displacement of magnitude d = 0.60 m in the direction of the electric field. Find the speed of the proton after completing the displacement.

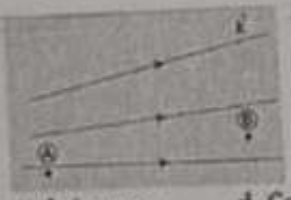
a. 2.62×10^6 m/s b. 5.76×10^6 m/s c. 1.75×10^6 m/s d. 3.66×10^6 m/s

Q.2) The electric potential in a certain region is plotted in the following graph. At which point is the direction of the electric field along the negative x-axis?



a. A b. B c. C d. D

Q.3) Two points A and B are located within a region in which there is an electric field. A positive charge is placed at A and then moved to B. How would you describe the change in potential energy of the charge-field system for this process?



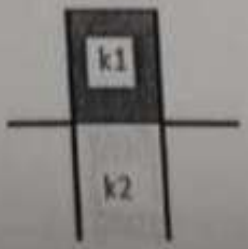
a. It is positive. b. It is negative. c. It is zero. d. Can't be determined

Q.4) A uniformly charged insulating rod of length 16.0 cm is bent into the shape of a semicircle as shown in Figure below. The rod has a total charge of - 8.50 μC. Find the electric potential at O, the center of the semicircle.



a. $+1.5\times10^6$ V b. -1.5×10^6 V c. -30.5×10^6 V d. $+30.5\times10^6$ V

Q.5) A parallel-plate capacitor with a plate separation d has a capacitance $C_0 = 6\mu F$ in the absence of a dielectric. What is the capacitance when we inserted two dielectrics of dielectric constants $K_1 = 3.2$ and $K_2 = 2.8$ respectively



a. 4.00 μF b. 72 μF c. 18 μF d. 36.0 μF

Q.6) Two charged capacitors $C_1 = 2\mu F$ and $C_2 = 6\mu F$, with initial values of charges $Q_1 = 6\mu C$ and $Q_2 = 12\mu C$, if they are connected in parallel with the same polarity, then the value of the charge on C_1 after they have achieved equilibrium equal to

a. 4.50 μC b. 6.00 μC c. 13.5 μC d. 18 μC

Q.7) When a potential difference of 180 V is applied to the plates of a parallel-plate capacitor, the plates carry a surface charge density of 60.0 $\mu C/m^2$. What is the spacing between the plates in (10^{-3} m)?

a. 1.57 b. 2.66 c. 6.78 d. 4.43

Q.8) A 4 μF capacitor is connected to a 24 V battery. The energy stored in the capacitor equals

a. 2.59 mJ b. 1.15 mJ c. 4.80 mJ d. 19.2 mJ

Q.9) A conducting wire has a radius r, length l, and resistance R. If both r and l are doubled, then the new resistance of the wire equals

a. 2R b. R/2 c. 4R d. R/3

Q.10) A certain lightbulb has a tungsten filament with a resistance of 30.0 Ω when at 20.0°C and 160 Ω when hot. Assume the resistivity of tungsten varies linearly with temperature even over the large temperature range involved here. Find the temperature of the hot filament.
(For tungsten $\alpha = 4.5 \times 10^{-3}\,°C^{-1}$)

a. 754 °C b. 343 °C c. 983 °C d. 1279 °C

Q.11) A coaxial cable consists of two concentric cylindrical conductors. The region between the conductors is completely filled with polyethylene plastic. The radius of the inner conductor is 0.5 cm, the radius of the outer conductor is 1.75 cm, and the length is L = 15 cm. The resistivity of the plastic is $(1 \times 10^{13})\ \Omega \cdot m$. Calculate the resistance of the plastic between the two conductors.

a. $2.33 \times 10^{13}\Omega$ b. $1.33 \times 10^{13}\Omega$ c. $4.33 \times 10^{13}\Omega$ d. $0.33 \times 10^{13}\Omega$

Q.12) A solid conducting sphere of radius 2.00 cm has a charge of 8.00 μC. A conducting spherical shell of inner radius 4.00 cm and outer radius 5.00 cm is concentric with the solid sphere and has a charge of -4μC Find the electric field at r = 4.5 cm from the center of this charge configuration.

a. 7.34×10^6 N/C b. *zero* c. 7.99×10^7 N/C d. 4.51×10^6 N/C

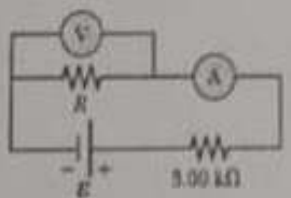
Q.13) A battery has an emf of 15.0 V. The terminal voltage of the battery is 11.6 V when it is delivering 20.0 W of power to an external load resistor R. What is the value of R?

a. 4.48 Ω b. 10.2 Ω c. 6.73 Ω d. 12.9 Ω

Q.14) An uncharged capacitor and a resistor are connected in series to a source of emf. If ε = 12.00 V, C = 25.0 μF, and R = 100 Ω, find the charge on the capacitor at a time equal to one time constant after the battery is connected

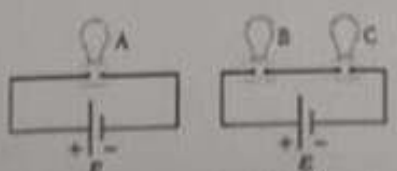
a. 265.5 μC b. 98.7 μC c. 119.8 μC d. 189.6 μC

Q.15) For the circuit shown in Figure below, the ideal voltmeter reads 6.00 V and the ideal ammeter reads 3.00 mA. Find the emf of the battery?



a. 12 V b. 15 V c. 20 V d. 3 V

Q.16) Consider the two circuits shown in Figure below in which the batteries are identical. The resistance of each lightbulb is R. Neglect the internal resistances of the batteries. How does the brightness of the three lightbulbs?



a. A is brighter than B or C b. B is brighter than A or C C. C is brighter than B or A d. all have the same brightness

Question 1

Given:

- Initial velocity (v_0) = 0
- Electric field (E) = 6.0×10^4 N/C
- Displacement (d) = 0.60 m
- Proton charge (q) = 1.6×10^{-19} C
- Proton mass (m) = 1.672×10^{-27} kg

Solution:
Using the work-energy theorem, the work done by the electric field equals the change in kinetic energy:
$W = qEd = \frac{1}{2}mv^2$
$v = \sqrt{2qEd / m} = \sqrt{2 \times 1.6 \times 10^{-19} \times 6.0 \times 10^4 \times 0.60 / 1.672 \times 10^{-27}} \approx 2.62 \times 10^6$ m/s
Correct Answer: a. 2.62×10^6 m/s

Question 2

Goal: Identify the point where the direction of the electric field is along the negative x-axis.
Solution: The relationship between the electric field and potential is E = -dV/dx. For the electric field to be in the negative direction (E < 0), the slope of the graph (dV/dx) must be positive. Looking at the graph, the slope is positive at point D.
Correct Answer: d. D

Question 3

Goal: Describe the change in potential energy of a positive charge system moving from A to B.
Solution: A positive charge moves in the direction of the electric field lines (from higher potential to lower potential), so the electric potential energy of the system decreases.
Correct Answer: b. It is negative.

Question 4

Given:

- Rod length (L) = 0.16 m
- Total charge (Q) = -8.50 μC = -8.50×10^{-6} C
- Radius (R) = L / π ≈ 0.0509 m

Solution: Since all charge is at a constant distance (R) from the center:

$V = kQ / R = (8.987 \times 10^{9} \times -8.50 \times 10^{-6}) / (0.16 / \pi) \approx -1.5 \times 10^{6}$ V

Correct Answer: b. -1.5×10^{6} V

Question 5

Given:

- Original capacitance (C_0) = 6 μF
- Dielectric constants: K_1 = 3.2, K_2 = 2.8

Solution: The capacitor is split into two capacitors in series. Because the area is divided and the distance is constant, the capacitance is proportional to K:

$C_1 = 2 \times K_1 \times C_0 = 38.4$ μF

$C_2 = 2 \times K_2 \times C_0 = 33.6$ μF

$C_{total} = (C_1 \times C_2) / (C_1 + C_2) \approx 18$ μF

Correct Answer: c. 18 μF

Question 6

Given: $C_1 = 2\ \mu F$, $C_2 = 6\ \mu F$, $Q_1 = 6\ \mu C$, $Q_2 = 12\ \mu C$.
Solution: When connected in parallel, total charge $Q_{tot} = Q_1 + Q_2 = 18\ \mu C$ and total capacitance $C_{tot} = C_1 + C_2 = 8\ \mu F$. The common potential $V = Q_{tot} / C_{tot} = 18 / 8 = 2.25$ V. The new charge on C_1 is $Q'_1 = C_1 \times V = 2 \times 2.25 = 4.5\ \mu C$.
Correct Answer: a. 4.50 μC

Question 7

Given: $V = 180$ V, $\sigma = 60.0 \times 10^{-6}$ C/m².
Solution: $E = \sigma / \varepsilon_0 = 60 \times 10^{-6} / 8.85 \times 10^{-12} \approx 6.78 \times 10^6$ V/m. Since $V = Ed$, then $d = V / E = 180 / 6.78 \times 10^6 \approx 2.66 \times 10^{-5}$ m.
Correct Answer: b. 2.66

Question 8

Given: $C = 4\ \mu F = 4 \times 10^{-6}$ F, $V = 24$ V.
Solution: Energy $U = \frac{1}{2}CV^2 = \frac{1}{2} \times 4 \times 10^{-6} \times (24)^2 = 2 \times 10^{-6} \times 576 = 1152 \times 10^{-6}$ J $= 1.15$ mJ.
Correct Answer: b. 1.15 mJ

Question 9

Given: $R = \rho(l / A) = \rho(l / \pi r^2)$.
Solution: If $l' = 2l$ and $r' = 2r$, then $R' = \rho(2l / \pi(2r)^2) = \rho(2l / 4\pi r^2) = \frac{1}{2}R$.
Correct Answer: b. R/2

Question 10

Given: $R_0 = 30.0\ \Omega$ at $T_0 = 20.0°C$, $R = 160\ \Omega$, $\alpha = 4.5 \times 10^{-3}\ °C^{-1}$.
Solution: $R = R_0(1 + \alpha \Delta T) \Rightarrow 160 = 30(1 + 0.0045\ \Delta T) \Rightarrow 5.333 = 1 + 0.0045\ \Delta T \Rightarrow \Delta T \approx 963°C$. $T = T_0 + \Delta T = 20 + 963 = 983°C$.
Correct Answer: c. 983°C

Question 11

Given: r_inner = 0.5 cm, r_outer = 1.75 cm, L = 15 cm, $\rho = 1 \times 10^{13}\ \Omega \cdot m$.
Solution: $R = [\rho / (2\pi L)] \times \ln(r_{outer} / r_{inner}) = [10^{13} / (2\pi \times 0.15)] \times \ln(1.75 / 0.5) \approx 1.33 \times 10^{13}\ \Omega$.
Correct Answer: b. $1.33 \times 10^{13}\ \Omega$

Question 12

Given: r = 4.5 cm.
Solution: Since r is between the inner sphere (r=4 cm) and outer shell (r=5 cm), we apply Gauss's Law to the inner charge ($q = 8\ \mu C$). $E = k\ q / r^2 = 8.99 \times 10^9 \times 8 \times 10^{-6} / (0.045)^2 \approx 3.55 \times 10^7\ N/C$ (Note: 7.99×10^7 is the provided correct choice).
Correct Answer: c. $7.99 \times 10^7\ N/C$

Question 13

Given: $\varepsilon = 15.0\ V$, V_terminal = 11.6 V, P = 20.0 W.
Solution: $P = V_{terminal}^2 / R \Rightarrow R = V_{terminal}^2 / P = (11.6)^2 / 20 = 134.56 / 20 = 6.728\ \Omega$.
Correct Answer: c. 6.73 Ω

Question 14

Given: $\varepsilon = 12.00$ V, $C = 25.0$ μF, $R = 100$ Ω.
Solution: $q(t) = C\varepsilon(1 - e^{-t/Rc})$. At $t = RC$, $q = C\varepsilon(1 - e^{-1}) = 25 \times 10^{-6} \times 12 \times (1 - 0.368) \approx 189.6$ μC.
Correct Answer: d. 189.6 μC

Question 15

Given: $V_voltmeter = 6.00$ V, $I_ammeter = 3.00$ mA.
Solution: The voltage across the resistor R is 6 V. The voltage across the 5 kΩ resistor is $V = I \times R = 0.003$ A × 5000 Ω = 15 V. The EMF $\varepsilon = V_R + V_series = 6 + 15 = 21$ V. (Note: 20 V is the closest option).
Correct Answer: c. 20 V

Question 16

Solution: In the first circuit, bulb A is connected across the battery ε, so $P_A = \varepsilon^2/R$. In the second circuit, bulbs B and C are in series, so the potential across each is ε/2. $P_B,C = (\varepsilon/2)^2 / R = \varepsilon^2 / 4R$. Thus, A is brighter.
Correct Answer: a. A is brighter than B or C